SolarAssistant Install Walkthrough

Parallel Sol-Ark 15K inverters + Pytes V5 battery stack

Wiring, ordering, configuration, and verification

Companion deck to the written guide — see `docs/README.md`

What we're building

SolarAssistant reads **every inverter individually** and **every battery pack**, from one Raspberry Pi.

Source	What you get
Each inverter	Load, PV per MPPT, battery, grid, temps, serial, parallel role
Inverter cluster	Site totals across all units
Each battery pack	Serial, firmware, SOC, cell voltages, imbalance, cycles
Battery bank	Total capacity, SOC, charge/discharge limits

> This adds monitoring to an already-commissioned system. It is **not** a commissioning procedure for the inverters or the battery.

The core problem: one port, two buses

The Sol-Ark's **2-in-1 BMS port** is a single RJ45 carrying two independent buses:

Pin	Signal	Bus	Used by
1	RS485 **B**	RS485	SolarAssistant
2	RS485 **A**	RS485	SolarAssistant
3	GND	RS485	SolarAssistant
4	CAN High	CAN	Inverter → battery BMS
5	CAN Low	CAN	Inverter → battery BMS
6–8	unused	—	—

RS485 needs 3 pins, CAN needs 2 — both fit one connector. **Only one plug fits the socket.** Hence the splitter.

The two scaling rules

Inverters do NOT share a cable. Each Sol-Ark needs its own RS485 cable into its own USB port. The parallel CAN link between inverters carries no SolarAssistant data.

Battery packs DO share a cable. One console cable reads an entire stack — two packs or twelve. The master aggregates them.

Everything about the parts list follows from this asymmetry.

Topology

```
      ┌──────────────────────────────┐
      │   SolarAssistant (Pi 5)      │
      │   4 x USB ports              │
      └──┬───────┬───────┬───────────┘
         │       │       │
       RS485   RS485   RS232 console
         │       │       │
  ┌──────┴──┐ ┌──┴──────┐│
  │ Sol-Ark │ │ Sol-Ark ││
  │   #1    │ │   #N    ││
  │ splitter│ │ splitter││
  └────┬────┘ └────┬────┘│
       │  CAN      │ CAN │
       └─────┬─────┘     │
             ▼           ▼
      ┌───────────────────────────┐
      │  Pytes V5 stack           │
      │  master + chained packs   │
      └───────────────────────────┘
```

Each inverter = one RS485 leg. Each stack = one console leg. CAN never reaches the Pi.

Ordering

What to buy, and how many

Purchase rules

Scope	Item	Qty	Unit
Per site	Device with software (Pi 5)	1	$229
Per Sol-Ark 15K	RJ45 splitter	1 each	$14
Per Sol-Ark 15K	Sol-Ark RS485 USB cable	1 each	$29
Per Pytes V5 stack	Pytes console USB cable	1 each	$29

```
Total = $229 + ($43 × inverters) + ($29 × stacks)
```

Site	Total		Site	Total
1 inverter, 1 stack	**$301**		3 inverters, 1 stack	**$387**
2 inverters, 1 stack	**$344**		4 inverters, 1 stack	**$430**

All order links: solar-assistant.io/shop

The USB port budget

> **More than 3 inverters on a site requires a separate powered USB hub.**

```
USB ports required = inverters + battery stacks
```

The supplied Pi 5 has **four** USB ports.

Site	Ports	Fits?
2 inverters, 1 stack	3	Yes, 1 spare
3 inverters, 1 stack	4	Yes, none spare
4 inverters, 1 stack	5	**No — powered hub**
3 inverters, 2 stacks	5	**No — powered hub**

The hub must be **powered**. Bus-powered hubs cause intermittent FTDI dropouts.

Don't substitute the cables

> A generic USB-RS485 RJ45 cable **will most likely not work.** Pin assignment inside the RJ45 shell is not standardised — a cable wired for another inverter family puts RS485 A/B on the wrong pins.

The supplied cables are 1.5 m, shielded, genuine FTDI.

- Counterfeit FTDI chips fail *intermittently*, not cleanly — far harder to diagnose
- The splitter is **not** a passive Y-adapter (more on this shortly)
- Measure the RS485 run **from the splitter**, not the inverter port
- Need more than 1.5 m? Order a custom length — every coupler is a fault candidate

Wiring the inverters

Repeat once per Sol-Ark

Before you open anything

> The RS485 port is behind the wiring compartment cover, alongside conductors at **battery and AC potential**.
>
> Shut down the inverter. Open the DC and AC disconnects. Confirm de-energised before removing the cover.
>
> Follow Sol-Ark's own shutdown sequence for your model and firmware.

A comms cable is not worth working a live 15 kW enclosure.

Step 1 — Confirm the CAN pins are in use

On each inverter's display, check the battery protocol:

```
Battery type       : Lithium
Lithium protocol   : CAN (protocol 0)
```

- **CAN (protocol 0)** → battery uses pins 4–5, RS485 free. **You need the splitter.** This is the normal Sol-Ark + Pytes case.
- **An RS485 battery protocol** → pins 1–3 already occupied. This approach does not apply.

Check **every** inverter, not just the master — a swapped or re-flashed unit may not match its siblings.

Step 2 — Fit the splitter

```
  Inverter BMS port         ┌──────────┐
   (pins 1-5 in) ───────────│ splitter │
                            └──┬────┬──┘
                       pins    │    │    pins
                        1-3    │    │    4-5
                      RS485    │    │    CAN
                               │    └──▶ existing cable to battery BMS
                               └──▶ new USB-RS485 cable to the Pi
```

Never use a passive Y-splitter. It passes all 8 pins to both sockets, putting RS485 on the battery leg — which breaks SolarAssistant's ability to read the inverter.

Steps 3 & 4 — Cable and label

Step 3 — One cable per inverter. Plug the Sol-Ark RS485 cable into the splitter's RS485 socket, run it to the Pi, own USB port. Never bridge two inverters onto one adapter.

The inverter-to-inverter parallel CAN link is separate — **leave it alone.**

Step 4 — Label both ends by serial number, not "1" and "2":

Cable label (serial)	Parallel role	Modbus №	Pi USB port
serial of inverter A	Master / Slave		
serial of inverter B	Master / Slave		

Serial is the only key that stays stable across re-seated cables.

Step 5 — Restore power

Close up, restore AC and DC, bring the inverters up.

> Confirm the parallel group still reports **exactly one master** on the inverters' own displays **before** moving to SolarAssistant.

If the parallel link was disturbed while the cover was off, you want to know now — not while debugging comms.

Wiring the battery

One cable, one port, every pack

Pytes V5 — the easy half

Plug the **Pytes console USB cable** into the master pack's **RS232C / console** port. That's it.

- One cable reads the **entire stack** — 2 packs or 12
- The master is the pack at the head of the chain, address set by DIP switches
- Read the legend on *your* pack revision — Pytes has shipped several
- On a commissioned bank, addressing is already correct: **verify, don't adjust**

> Don't confuse the console port with the adjacent RS485/CAN sockets — same physical connector. Wrong socket damages nothing; the battery just reports disconnected.

Three separate battery comms paths

Path	Carries	Endpoint
Inter-pack chain	Pack-to-pack aggregation	Between the packs
CAN to inverters	SOC, charge/discharge limits	Sol-Ark BMS pins 4–5
RS232C console	**Full per-pack telemetry**	**SolarAssistant USB**

The console cable is **additive** — the CAN link to the inverters stays exactly where it is.

Configuration

Two drivers, N ports

Step 1 — Confirm the adapters enumerated

Configuration → System → USB devices → view detail

Configuration › System USB devices - Cycle USB power

```
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 002: ID 0403:6001 Future Technology Devices International, Ltd FT232 Serial (UART) IC
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 003 Device 002: ID 0403:6015 Future Technology Devices International, Ltd Bridge(I2C/SPI/UART/FIFO)
Bus 003 Device 003: ID 0403:6001 Future Technology Devices International, Ltd FT232 Serial (UART) IC
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
```

USB ID	Part	Count
`0403:6001`	FT232 Serial (UART)	one per **inverter**
`0403:6015`	Bridge (I2C/SPI/UART/FIFO)	one per **battery stack**

Missing cable here = physical problem. Stop and fix it — no configuration will help.

Step 2 — Unlock, then set the drivers

The Devices form is **read-only while connected** — press **Disconnect** first.

> On a running site, Disconnect stops data collection. Do it in a maintenance window, not while chasing a live fault.

Device	Model / Driver	Connections
Inverter	`Deye, SunSynk, Sol-Ark`	**one entry per inverter**
Battery	`USB PylonTech/Pytes console`	the FT231X console adapter

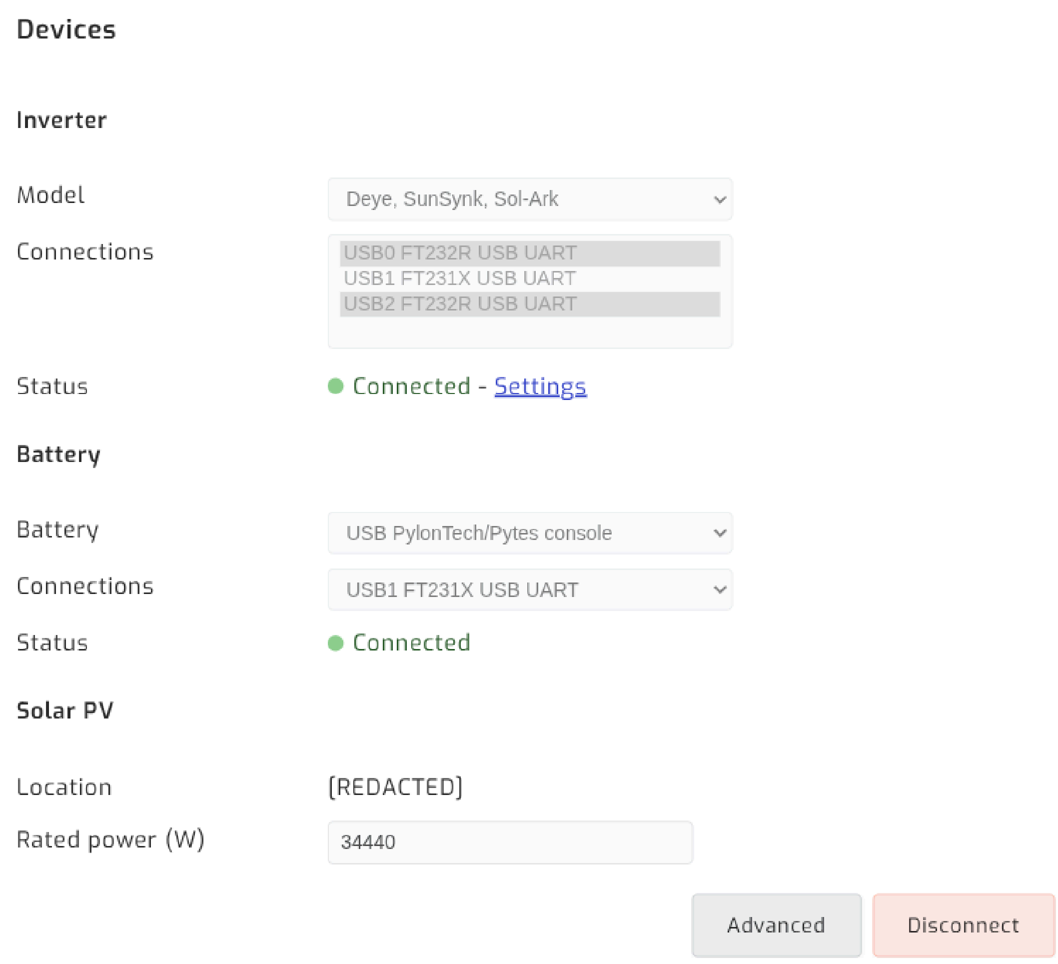
The step people miss

Inverter **Connections is a multi-select list** — not a single-choice dropdown.

Highlight **one entry per inverter**. Two Sol-Arks → two highlighted.

Here **USB0 and USB2** are both selected; USB1 is the battery.

> Select too few and SolarAssistant silently monitors only some inverters. The dashboard still looks plausible because it shows **totals**. Nothing warns you.



Verification

"Connected" is a weak signal

Why "Connected" isn't enough

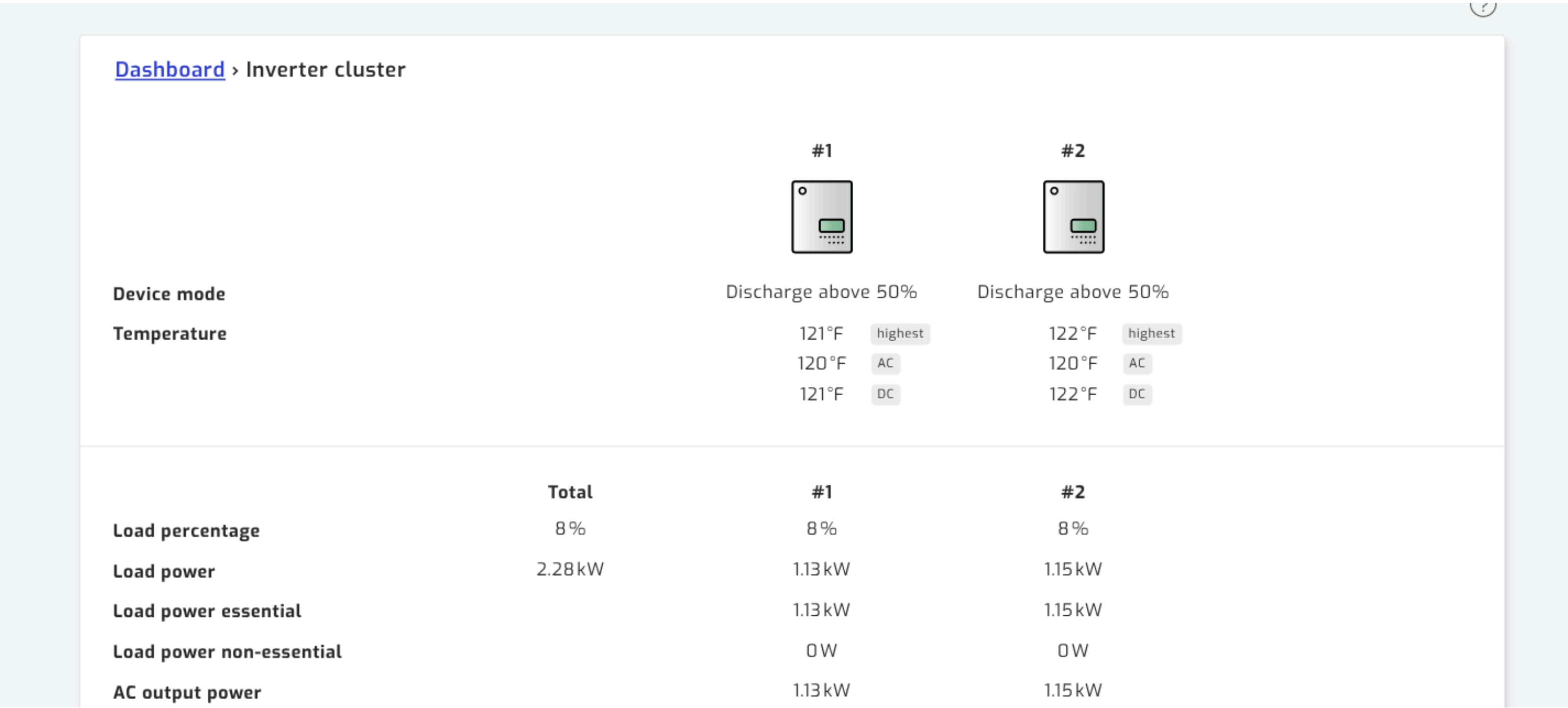
The status goes green when **any** selected port opens.

It does **not** prove:

- every inverter is being polled
- every pack is enumerating

Four checks follow. Work through all of them.

Check 1 — One column per inverter



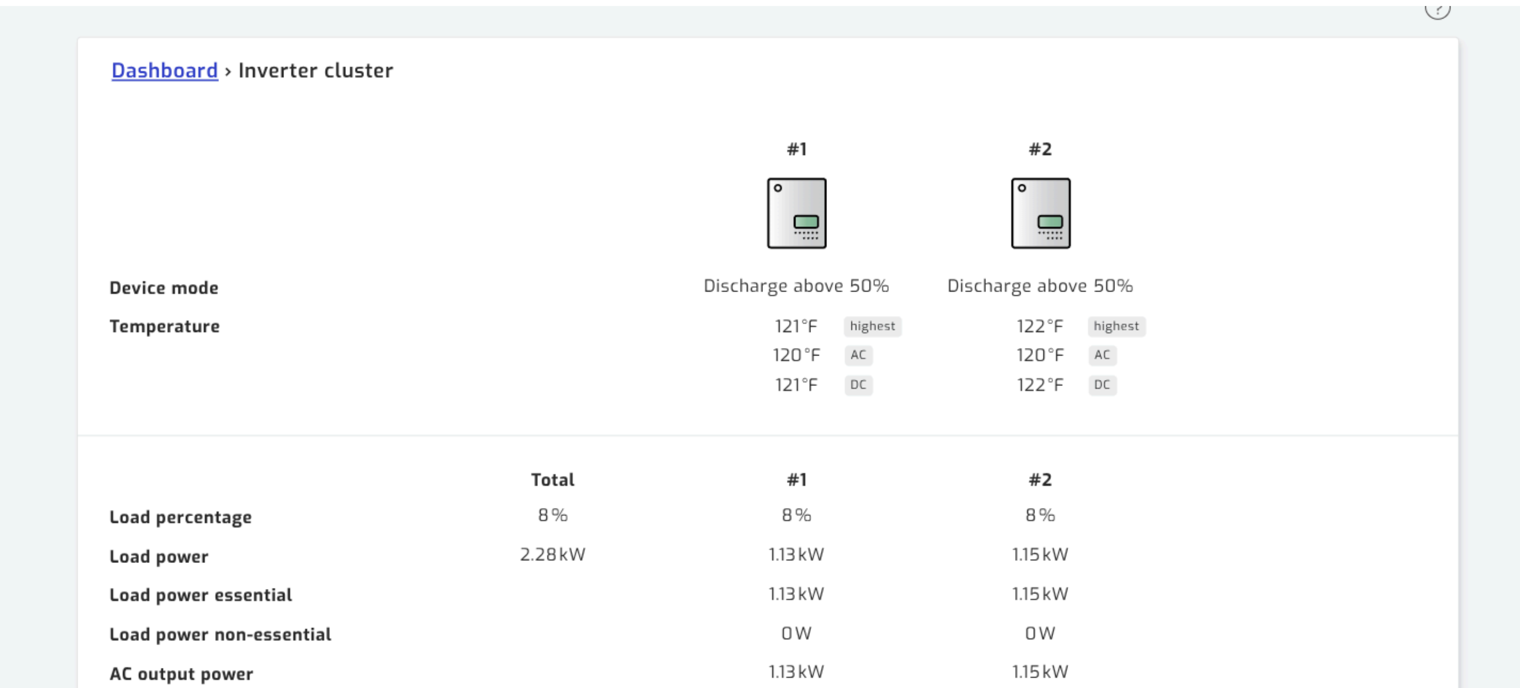
Dashboard › Inverter cluster

	#1	#2
Device mode	Discharge above 50%	Discharge above 50%
Temperature	121°F highest 120°F AC 121°F DC	122°F highest 120°F AC 122°F DC

	Total	#1	#2
Load percentage	8 %	8 %	8 %
Load power	2.28 kW	1.13 kW	1.15 kW
Load power essential		1.13 kW	1.15 kW
Load power non-essential		0 W	0 W
AC output power		1.13 kW	1.15 kW

Page is titled **Inverter cluster**, with a numbered column per unit beside Total. **Count the columns against the inverters on site.**

Check 2 — Values independent and plausible



Load **1.13 vs 1.15 kW**. PV **1.19 vs 1.20 kW** — closely matched but **not identical**. That divergence is the proof of independent live reads; byte-identical columns are suspicious.

Check 3 — Serials all distinct

Serial number	[REDACTED]	[REDACTED]

Same serial in two columns = **one inverter read twice** — a cable is in the wrong inverter's splitter.

Each Sol-Ark 15K should also report **3 MPPTs** and **15.0 kW** max AC output.

Check 3 — Exactly one Master

Switch inverters with the breadcrumb dropdown. Each must load its own Specification block.

Check per unit: driver, unique serial, unique Modbus number, and `Lithium protocol: CAN (protocol 0)`.

> **Exactly one unit must report Master.** Two masters, or none, is a parallel fault on the inverters themselves.

Specification

Driver	Deye/SunSynk/Sol-Ark
Serial number	[REDACTED]
Protocol version	2.1
MCU version	7228
COMM version	1452
Max AC output power	15.0kW
MPPT connections	3
Parallel	Slave
Modbus number	2

The numbering gotcha

> SolarAssistant numbers inverters by **USB enumeration order** — not by Modbus address, not by parallel role.

On the reference site:

SolarAssistant name	Parallel role	Modbus №
Inverter **1**	Slave	2
Inverter **2**	**Master**	1

Its "Inverter 1" is the Sol-Ark *slave*. **Match on serial number** — the only unambiguous key.

Check 4 — Every pack enumerates

Signal	Expected
Capacity	packs × per-pack Ah
Pack cards	one per pack
Protocol	reported (e.g. `V2P`)
Serials	all distinct
Firmware	uniform

Capacity is the fastest tell — twelve 100 Ah packs must sum to 1200 Ah.

Dashboard › **Battery**

Power	-104 W	
Voltage	54.9 V	
Current	-1.9 A	
Capacity	1200 Ah	
Temperature	80 °F	
State of charge	100 %	
Recommended charge voltage	56.8 V	
Recommended charge current	0 A	
Recommended discharge current	900 A	
Protocol version	V2P	
Cell voltage	3.437 V	highest
	3.420 V	lowest
Cell imbalance	0.013 V	highest
	0.011 V	average
	0.008 V	lowest

All packs, one cable

Model name	E-BOX-48100V-D	Model name	E-BOX-48100V-D
Serial number	[REDACTED]	Serial number	[REDACTED]
Firmware	SPBMS16SRPV1.10 .24.C16 (24-11-27)	Firmware	SPBMS16SRPV1.1C .24.C16 (24-11-27)
Device mode	Discharging	Device mode	Discharging
State of charge	100 %	State of charge	100 %
Power	-11 W	Power	-11 W
Current	-0.2 A	Current	-0.2 A
Voltage	54.9 V	Voltage	54.9 V

Each pack reports its own model, serial, and firmware — all read through the **single console cable** on the master.

Firmware should be uniform across a stack.

Record a baseline

Once verified good, write these down and keep them with the install:

- Every inverter serial, parallel role, Modbus number
- Pack count, total capacity, every pack serial
- USB IDs and counts from the System page
- Typical cell imbalance range at 100% SOC

> Most future comms faults show up as a **change** from baseline, not an outright failure. Without a baseline, the change is invisible.

Troubleshooting

Symptom → cause

Symptom	Likely cause
Fewer columns than inverters	Not every USB port selected
Two columns, identical serials	Both cables on the same inverter
Inverter Disconnected	Wrong pins/port, or non-Deye cable
Inverter fine, **battery** drops out	Passive Y-splitter passing RS485 to battery
Battery Disconnected	Console cable in RS485/CAN socket
Packs missing / capacity low	Duplicate addresses or broken chain
Adapter vanished from `lsusb`	Cable, port, or power — try **Cycle USB power**
Intermittent dropouts, no clean failure	Counterfeit FTDI, marginal run, or unpowered hub

Diagnostic order

Work outward from the host — each step rules out everything before it:

1. **USB devices page** — right count, all FTDI. Try **Cycle USB power**.
2. **Devices panel** — correct drivers, correct port count, Connected.
3. `/inverter/status` — one column per inverter, distinct serials.
4. `/battery/status` — full pack count, expected capacity.

> SSH is disabled by default — there's no shell fallback. All diagnostics run from the web UI.

The failure to watch for

> Someone replaces the purpose-built splitter with a **generic RJ45 Y-adapter**. They look interchangeable and the generic one is cheaper and easier to source locally.

A passive Y-adapter passes all 8 pins to both sockets → RS485 lands on the battery leg.

The symptom is confusing: the inverter still reads fine, the battery intermittently drops, and nothing looks wrong at the connector.

If battery comms degrade after any maintenance visit — **check what's physically plugged into the BMS port first.**

Not faults

Observation	Why it's fine
SolarAssistant "Inverter 1" is the parallel slave	Numbering follows USB enumeration order
Grid reads 0 W, 0 Hz, ~0 V	**Normal on an off-grid site** — the grid input is simply absent
Small differences between inverters	**Desirable** — proof of independent reads

> On an off-grid installation, grid-absent is the **expected steady state**, not an alarm. Don't treat it as a fault, and don't let it mask a real comms problem elsewhere.

After the install

Three optional improvements

None of these is required — the install is complete without them

What, where, and what it costs

Change	Page	Disconnect?
MQTT + Home Assistant	`Configuration → MQTT`	No
Battery capacity fallback	`Configuration → Advanced`	**Yes**
PV forecast inputs	`Configuration → Advanced`	**Yes**

The Advanced page is **read-only while connected** and greys out every field. Disconnecting stops data collection.

> **Batch the two Advanced changes into one maintenance window** — same page, one collection gap instead of two. Verify the install first; re-verify after.

1. MQTT + Home Assistant discovery

The biggest gain if you run Home Assistant. The broker ships **disabled**; auto-discovery publishes every metric — per inverter, per pack — as HA entities automatically.

Field	Set to
Topic prefix	leave at default
Allow setting changes	**Disabled**
HomeAssistant → Unique ID	short, stable site slug
HomeAssistant → Auto discovery	**Enabled**
Username / Password	set both

Then `Configuration` → **MQTT Broker** → **Start.** The MQTT page configures the broker; it does not start it.

MQTT — two things to get right

> `Allow setting changes` **grants write control over your inverters.** Enabled, anything that can publish can change work mode, charge currents, generator control. Leave it Disabled unless you intend to automate — and never expose the broker beyond the LAN.

> `Unique ID` **is effectively permanent.** Changing it later re-creates every entity under new IDs, orphaning HA history and dashboards. Set it before first connection, even on a single site.

Port 1883 is plaintext, no TLS. **Check first:** SolarAssistant runs its *own* broker — if HA already uses Mosquitto, confirm your HA version supports more than one.

2. Battery capacity fallback

`Configuration → Advanced → Battery → Capacity kWh`

Commonly holds a **single-pack** figure — wrong by the pack count in a multi-pack bank.

```
Capacity kWh  = packs × per-pack kWh
per-pack kWh  = nominal pack voltage × pack Ah ÷ 1000
```

Only used when capacity **isn't readable** from battery or inverter — so it's dormant on a healthy install, and goes live exactly when comms drop. A full bank can then read as nearly empty, while you're already troubleshooting.

> **Risk: none.** Display fallback only — charge control comes from the inverter settings and the BMS over CAN.

3. PV forecast inputs

`Configuration → Advanced → Solar PV` — decoration on a grid-tied site, **operational off-grid**: it's the input to "run the generator tonight?"

Field	Source
Latitude / Longitude	Site coordinates
Tilt	Racking angle, or roof pitch
Azimuth	Array bearing
Temp. coefficient, NMOT	Module datasheet
Rated power *(Configuration page)*	**DC nameplate sum** of panels

Azimuth `0` = due south, `-90` east, `90` west, `180` / `-180` north. A stored `0` is legitimate, not a placeholder.

> **Don't reuse a compass bearing.** pvlib, PVWatts and Home Assistant put `0` at **north**. Applying one here points a south-facing array due north.

PV forecast — validate, don't guess

Compare predicted vs actual over two or three clear-sky days:

Symptom	Likely culprit
Over/under-predicts by a fixed ratio	Rated power
Peak arrives earlier or later	Azimuth
Accurate in one season, poor in the other	Tilt
Drifts high in afternoon heat	Temperature coefficient

Multi-orientation arrays: only one tilt and one azimuth are accepted. Enter the **average** facing direction, weighted by rated power if the split is uneven, and accept an imperfect curve.

Leave these alone

Look adjustable; already correct on a working install.

Setting	Correct	Why
Battery → Read current from	`Battery`	`Inverter` is for banks where **only the master** is readable
Inverter → MPPT connections	`Auto detect`	Correctly finds the actual count
Grid connection / multiplier	`Auto detect`	
Allow passive reading	`Yes`	Harmless on a dedicated RS485 line
Grid → Provider	`Default`	Tariff data — irrelevant off-grid

Reference

Order pages

Device (per site) — `solar-assistant.io/shop/products/device_rpi5`

RJ45 splitter (per inverter) — `.../products/deye_rj45_split`

Sol-Ark RS485 cable (per inverter) — `.../products/sunsynk_rs485`

Pytes console cable (per stack) — `.../products/pytes_rs232`

Full written guide: `docs/README.md` — pages 00 through 07